

Advanced (Habanero)

- Q1.** What is the unit rate of $15 \div 3$?
(A) 3
(B) 5
(C) 4
(D) 2
(E) 1
- Q2.** A car travels 250 miles in 5 hours. What is the unit rate?
(A) 50 miles/hour
(B) 40 miles/hour
(C) 45 miles/hour
(D) 55 miles/hour
(E) 60 miles/hour
- Q3.** If y varies directly with x , and $y = 12$ when $x = 4$, find the constant of variation.
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** A worker can build 8 chairs in 2 hours. How many chairs can the worker build in 5 hours?
(A) 10 chairs
(B) 16 chairs
(C) 20 chairs
(D) 24 chairs
(E) 30 chairs
- Q5.** If y varies inversely with x , and $y = 6$ when $x = 2$, find y when $x = 3$.
(A) 3
(B) 4
(C) 6
(D) 8
(E) 10
- Q6.** A plane travels 480 miles in 8 hours. What is the unit rate?
(A) 50 miles/hour
(B) 55 miles/hour
(C) 60 miles/hour
(D) 65 miles/hour
(E) 70 miles/hour
- Q7.** If z varies directly with the square of x , and $z = 36$ when $x = 3$, find z when $x = 6$.
(A) 80
(B) 100
(C) 144
(D) 160
(E) 200
- Q8.** A cyclist travels 25 miles in 2 hours. How many hours will it take the cyclist to travel 50 miles?
(A) 2 hours
(B) 3 hours
(C) 4 hours
(D) 5 hours
(E) 6 hours

Open Ended Questions (FRQ)

- Q9.** A bakery sells 480 loaves of bread in 8 days. If the bakery operates 6 days a week, how many loaves of bread will it sell in 5 weeks? Show your work and explain your reasoning.
- Q10.** The cost of producing x units of a product is given by the equation $C(x) = 2x + 100$. The revenue from selling x units is given by $R(x) = 5x - 0.1x^2$. Find the break-even point, where $C(x) = R(x)$. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Make sure to read the questions carefully and choose the correct units.
- Tip 2: Use the given information and formulas to solve the problems.
- Tip 3: Check your work and explain your reasoning for FRQs.